

COMPUTER SCIENCE TRIPOS Part IB 75%, Part II 50% – 2021 – Paper 7

6 Further Graphics (aco41)

Surface
Representations -
Implicit Functions

- (a) An implicit function based on a kernel regressor can be defined as $f(\mathbf{x}) = \sum_i \alpha_i k_i(\mathbf{x})$, where α_i are scalar coefficients, $k_i(\mathbf{x}_i) = e^{-\|\mathbf{x}-\mathbf{x}_i\|^2/\sigma^2}$ are kernels, and $\mathbf{x}_i$ are sample points.
- (i) Can $\alpha_i > 0$ define a valid implicit function, why/why not? [1 mark]
- (ii) For a point $\mathbf{x}$ on the surface, compute the surface normal, simplify as much as possible. [2 marks]
- (iii) Assume the points $\mathbf{x}_i$ are sampled from a plane passing through the origin. Will the regressor based implicit function preserve the plane normal? Show it mathematically. If not, explain a solution to better approximate the plane with the kernel regressor in 1-2 sentences. [3 marks]

Answer:

- (i) It cannot as $f(\mathbf{x}) > 0$ for all $\mathbf{x}$ in that case.
- (ii) $\nabla f(\mathbf{x}) = 2/\sigma^2 \sum_i \alpha_i \mathbf{x}_i k_i(\mathbf{x})$ by direct differentiation and using the fact that $f(\mathbf{x}) = 0$. This should be normalized, giving $\mathbf{n}(\mathbf{x}) = \sum_i \alpha_i \mathbf{x}_i k_i(\mathbf{x}) / \|\sum_i \alpha_i \mathbf{x}_i k_i(\mathbf{x})\|$.
- (iii) As $\mathbf{x}_i$ are on a plane passing through the origin, for the plane normal $\mathbf{n}$, $\mathbf{n}^T \mathbf{x}_i = 0$. Starting from the gradient of the regressor based implicit function we can write

$$\left(\sum_i \alpha_i (\mathbf{x} - \mathbf{x}_i) k_i(\mathbf{x}) \right)^T \mathbf{n} = \sum_i \alpha_i \mathbf{n}^T (\mathbf{x} - \mathbf{x}_i) k_i(\mathbf{x}) = 0.$$

Hence, the plane normal is not preserved. To solve this problem, we need off-surface sample points at which the kernels are defined.

Surface
Representations -
Implicit and
Parametric
Functions

- (b) We have two surfaces A and B , both of which are represented continuously. We would like to compute if they intersect. How would you represent A and B (parametric and/or implicit) and why? Explain in 1-2 sentences. [2 marks]

Answer: One of them can be a parametric surface and the other an implicit surface so that we can generate points from one and test whether they are inside or not by simply computing the implicit function for the other.

Differential
Geometry

- (c) (i) Prove that a plane has zero mean curvature by utilizing $\nabla_{\mathcal{M}} \mathbf{p} = -2H\mathbf{n}$. [2 marks]
- (ii) Given a triangular mesh with equilateral triangles and a vertex with non-negative discrete minimum and maximum curvatures, what is the maximum number of neighbours it can have? [3 marks]

Answer:

- (i) $\mathbf{p} = [x, y, z]^T$ and $\nabla_{\mathcal{M}} = \nabla$ for a plane. Hence, $\nabla_{\mathcal{M}} \mathbf{p} = \nabla[x, y, z]^T = 0$.

— *Solution notes* —

- (ii) The surface at the vertex given has positive or zero Gaussian curvature and hence $2\pi - \sum_j \theta_j \geq 0$. Since the triangles are equilateral, all θ_j are the same and equal to $\pi/3$. Thus we get $2\pi \geq n\pi/3$, $6 \geq n$.
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Transformation
Blending

- (d) Assume we embed a number of bones inside the sphere of radius 1 and with center at the origin and start trying to deform the sphere by only having rotations at the bones.

- (i) If we use linear blend skinning, show whether the transformed points can be off the sphere. [1 mark]
- (ii) If we use linear quaternion blending with normalization, show whether the transformed points can be off the sphere. [2 marks]
- (iii) If we use linear quaternion blending with normalization, will the sphere stay intact? Briefly explain. [2 marks]
- (iv) If we use linear quaternion blending with normalization, but this time the sphere we are deforming is centered at $[1, 1, 1]^T$, show whether the transformed points can be off the sphere. [2 marks]
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Answer:

- (i) The sphere is given by $\|\mathbf{x}\| = 1$. Linear blend skinning maps each of these points to $\sum_i w_i(\mathbf{x}) \mathbf{R}_i \mathbf{x} = \sum_i w_i(\mathbf{x}) \mathbf{x}_i(\mathbf{x})$, where $\mathbf{x}_i(\mathbf{x})$ are rotated versions of the initial point on the sphere and hence they are also on the sphere. A linear combination of those points does not necessarily reside on the sphere.
- (ii) Linearly blended and normalized quaternions will necessarily provide valid rotations. Hence, each point will be rotated around the origin and will still stay on the sphere.
- (iii) It may not. It can happen that a part of the sphere folds onto another part.
- (iv) In this case, as the rotation is around the origin and the sphere is translated, the points can move off the sphere.
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